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Low-heat alkaline pretreatment of biomass for dairy anaerobic codigestion

GUANG JIN and TOM BIERMA

Department of Health Sciences, Illinois State University, Normal, Illinois, USA

In this research, low-heat alkaline pretreatment was evaluated to determine the extent to which urban landscape waste (yard waste), corn stover, and switchgrass could be codigested under conditions typical of US farm-based anaerobic digestion (AD). Waste heat from combined heat and power (CHP) units associated with AD could make such pretreatment economical. Short-term batch digestion studies and 8-week continuous-feed studies were used to screen and evaluate various pretreatment conditions. Results indicate that maple and oak leaves did not digest well, even with pretreatment. Pretreatment did improve digestion of corn leaves and stalks as well as switchgrass. However, these materials also digested reasonably well even without pretreatment. No digester operational problems were observed during continuous-feed studies of intermittently stirred bench top digesters, but optimal levels of alkali, temperature, and pretreatment time may be specific to the feedstock, particle size, and digester loading rate. Results suggest that some common lignocellulosic biomass materials, such as corn stover and switchgrass, could be successfully codigested in many existing farm-based digesters. Interestingly, without pretreatment, switchgrass digestion improved over 20-fold when digested with seed culture from a dairy digester compared to seed culture from a municipal digester, suggesting that culture acclimation could be as important as pretreatment in improving digestion of specific lignocellulosic feedstocks.

Keywords: Anaerobic digestion, low-heat alkaline pretreatment, corn stover, maple leaves, oak leaves.

Introduction

Anaerobic digestion (AD) offers many benefits as a treatment technology for manure, including dramatic reduction of pathogens, biological oxygen demand (BOD), and odors, and the production of renewable energy in the form of methane.^[1,2] However, in the United States, only a little over 200 farm-based digesters are in operation.^[3] The primary barriers to AD adoption in the United States are generally considered to be high capital investment costs and low energy prices, resulting in unacceptably low returns on investment.^[4] In contrast to many European countries, where farm-based AD has been widely adopted due to policies supporting high returns from renewable energy supply, AD systems in the United States generally receive small or no price premiums over the local energy prices.^[4–6]

To supplement limited energy revenues, many digester operators have explored codigestion of manure with other

degradable organic materials. Codigestion can provide the benefit of increasing methane production and stabilizing digester operation by complementing the manure with high-carbon feedstocks, bringing the balance of carbon with nitrogen and other nutrients closer to optimal for AD operation.^[7] If the codigestion feedstock can be obtained at low cost relative to the value of the additional methane produced, codigestion has the potential to substantially improve the economic performance of AD.

Food processing wastes are generally considered the most desirable codigestion feedstock as they are often readily hydrolyzed, but availability of such wastes is geographically limited.^[8] In European digesters designed for codigestion, such as two-phase, continuously stirred tank reactors (CSTR) or “dry” digesters, more recalcitrant lignocellulosic feedstocks, such as corn stover, have been used.^[9] However, in the United States and other parts of the world, many farm-based digesters have not been specifically designed for codigestion and include many designs, such as plug flow and covered lagoon, that have relatively poor mixing and operate with a solid content under 12%.^[10] It is unclear whether lignocellulosic feedstocks could be codigested successfully in such digesters.

Many lignocellulosic feedstocks have been suggested for codigestion with manure, including switchgrass, corn stover, yard waste, waste paper, and even wood wastes.^[9,11]

Address correspondence to Guang Jin, Department of Health Sciences, Illinois State University, Normal, IL 61790, USA; E-mail: gjin@ilstu.edu

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In the United States, the lignocellulosic materials most widely available would be corn stover and yard waste. The US Environmental Protection Agency (US EPA) estimates the volume of yard waste generated annually to be about 34 million tons.^[12] Because many states have banned yard waste from landfills, approximately 57% (19 million tons) is currently composted or land-applied while the remainder goes to disposal. Tipping fees may be available to AD operators for accepting these wastes. Corn stover, on the other hand, would be available only at a cost. However, stover harvesting costs may decline as the cellulosic ethanol industry drives innovation in harvesting technologies and methods.^[13,14]

Acreage of energy crops in the United States is currently limited, but may grow as cellulosic ethanol technologies become more economical and widespread. Current estimates for the farm gate price needed to support switchgrass cultivation on marginal land range from about \$40 to \$100 ton⁻¹ depending on field productivity.^[15,16] Storage and transport would be expected to add to this cost.^[17]

Many investigators have examined pretreatment technologies to improve digestion of lignocellulosic materials. The availability of waste heat from combined heat and power (CHP) systems operating at many digesters creates the potential for relatively inexpensive thermal pretreatment.

The primary challenge in codigesting many biomass materials is their lignocellulosic content. Lignocellulose resists hydrolysis and enzymatic attack, making it difficult to digest in AD systems.^[18] The chemical composition of lignin; the matrix linking lignin, hemicellulose, and cellulose; and the crystalline structure of cellulose are believed to contribute to the recalcitrance of lignocellulosic feedstocks.^[19]

To overcome this recalcitrance, a number of pretreatment methods have been explored for lignocellulosic biomass prior to digestion. Both acids and bases have been shown to be effective in pretreatment.^[18–20] Dilute acids – most commonly sulfuric acid – appear to work primarily by preferentially hydrolyzing and removing hemicellulose, leaving cellulose more accessible in the residual solids.^[21] Alkaline pretreatment is considered one of the most promising options for pretreatment.^[22] Alkaline pretreatment appears to act primarily from degradation of the hemicellulose–lignin bonds with partial removal of the lignin from the biomass, increasing porosity, and exposing the cellulose and hemicellulose to subsequent enzyme attack.^[18] Compared to acid pretreatment, alkaline pretreatment can generally be performed under milder conditions (even room temperature), though pretreatment times are often hours or days compared to minutes for acid pretreatment.^[18]

The common use of AD-generated methane for electricity production creates the possibility of using waste heat from CHP systems to provide the heat needed for alkaline pretreatment, reducing the amount of alkaline required

and the overall cost of pretreatment. Our previous research demonstrated that mild heat (100°C) and moderate alkaline concentrations could be effective in improving switchgrass codigested in municipal wastewater treatment AD.^[23]

In the research reported here, we explore whether similar pretreatment conditions could be effective in improving the digestion of a number of lignocellulosic feedstocks under conditions typical of US farm-based AD systems. Specifically, we examine the seed culture, mixing regime, and residency time of plug flow AD using dairy manure as a feedstock. Lignocellulosic feedstocks evaluated are corn stover (leaves and stalks), yard waste (specifically oak and maple leaves), and switchgrass.

Materials and methods

Short-term batch digestion studies were used to screen a number of pretreatment conditions for maple and oak leaves and for corn leaves and stalks. The most promising pretreatment conditions were then used in longer-term continuous-feed studies to evaluate potential operational problems and better assess digestibility under steady-state conditions. Switchgrass was included in the continuous-feed studies based on results from prior research in batch studies.^[23]

Origin of materials

Red maple (*Acer rubrum*) and red oak (*Quercus macrocarpa*) are common tree species found in Midwestern US communities and could be anticipated to comprise a significant proportion of yard waste in autumn. Leaves were collected from the ground at a Midwestern university campus in November and air dried to an average moisture content of 4.82% and a volatile solid percentage (VS%) of 95.34% for maple and moisture content of 4.89% and a VS% of 95.47% for oak. Moisture and VS% reported for biomass used in this study are analyzed according to Standard Method 2540-B and Standard Method 2540-E, respectively. Both types of leaves were then prepared as either fine-grind or coarse-grind. To prepare fine-grind, leaves were ground in a Wiley Mill (Thomas Scientific, Swedesboro, NJ, USA) with a 2-mm screen. To prepare coarse-grind, leaves were ground in a Ronson Model HB-2 Blender (Ronson Consumer Products Corporation, Somerset, NJ, USA) until particles were small enough to fall through the bottom screen of a two-sieve Forage Particle Separator (NASCO, Modesto, CA, USA), resulting particles generally less than 0.78 cm. Coarse-grind and fine-grind maple and oak leaves are quite uniform in shapes and sizes.

Corn leaf and stalks were collected from the university farm in November. These were air dried to an average



Fig. 1. Coarse grind corn stover.

moisture content of 9.26% and a VS% of 94.29%. Corn leaves and stalks were only prepared as coarse-grind. To prepare coarse-grind, corn leaves and stalks were first cut into 2-inch pieces and then prepared in the same manner as coarse-grind oak and maple leaves, above. However, particles comprised a variety of shapes (see Fig. 1) and some particles were significantly larger than 0.78 cm in length. A corn stover mix was produced by combining coarse-grind leaves and stalks at a 1-to-1 ratio.

Cave-in-Rock switchgrass (*Panicum virgatum*) was obtained from the Horticulture Center at Illinois State University. Switchgrass was harvested in June (at the early flowering stage). It was air dried to an average moisture content of 4.74% and a VS% of 92.62%. Switchgrass was cut into 2-inch pieces using a paper cutter and then prepared in the same manner as coarse-grind oak and maple leaves, above. Seed sludge used to inoculate the bench-top digesters was effluent from a local dairy digester operating at 39°C (105°F).

Pretreatment conditions

Maple leaves received a variety of pretreatment conditions for batch studies (Table 1). Minimal pretreatment of

Table 1. Summary of pretreatment conditions for maple leaves.

Pretreatment code	NaOH conc. (wt/wt %)	Heating conditions	Rinse after pretreatment
Control ^a	NA	NA	NA
A	0%	25°C (77°F), 18 h	No
B	3%	100°C (212°F), 6 h	No
C	6%	100°C (212°F), 24 h	No
D	15%	100°C (212°F), 24 h	No
E	15%	100°C (212°F), 24 h	Yes
F	6%	Autoclave 121°C, 15 psi, 15 min	No
G	15%	Autoclave 121°C, 15 psi, 15 min	No
H	15%	Autoclave 121°C, 15 psi, 15 min	Yes

^aControl has no maple leaves, only feed sludge was added.

soaking with reverse osmosis (RO) water under room temperature was evaluated as a base condition (Condition A). Various concentrations of NaOH with mild heat (100°C or 212°F) for various heating hours were examined (conditions B–E). Note that all alkaline concentrations are expressed on a wt/wt (alkaline/biomass) basis. Autoclaving, producing an elevated pressure and higher temperature, was also evaluated (conditions F–H). The effects of rinsing pretreated maple prior to digestion (condition E and H) also were evaluated.

Based upon the results from batch tests of maple leaves, the most promising pretreatment conditions were selected for batch studies of oak leaves (Table 2). In addition, fine-grinding as a physical pretreatment condition was also evaluated. Pretreatment conditions evaluated in batch studies for corn leaves and stalks are presented in Table 3. Leaves and stalks were tested separately for individual digestibility.

To pretreat the biomass, 10.0 g of dry biomass was soaked in 60 mL NaOH solution (or RO water) in a beaker. The beaker was placed in a 100°C water bath or autoclave for the desired amount of time. When pretreatment was complete, biomass was drained of liquid using a strainer for 5 min (no drying). If rinsing was specified, a spray bottle of RO water was used to administer a 5-s rinse

Table 2. Summary of pretreatment conditions for oak leaves.

Pretreatment code	NaOH conc. (wt/wt%)	Heating conditions	Rinse after pretreatment	Type of grind
Control ^a	NA	NA	NA	NA
A	0%	25°C (77°F), 18 h	No	Coarse
B	3%	100°C (212°F), 24 h	No	Coarse
C	3%	Autoclave 121°C, 15 psi, 15 min	No	Coarse
D	NA	NA	NA	Fine
E	6%	25°C (77°F), 18 h	No	Fine
F	6%	100°C (212°F), 24 h	No	Fine

^aControl has no oak leaves, only feed sludge was added.

Table 3. Summary of pretreatment conditions for corn leaves and stalks.

Pretreatment code	NaOH conc. (wt/wt%)	Heating conditions	Rinse after pretreatment	Type of corn stover
Control ^a	NA	NA	NA	NA
A	3%	100°C (212°F), 6 h	No	Corn leaf
B	3%	Autoclave 121°C, 15 psi, 1 h	No	Corn leaf
C	4%	100°C (212°F), 6 h	No	Corn stalk
D	4%	Autoclave 121°C, 15 psi, 1 h	No	Corn stalk

^aControl has no corn leaves or stalks, only feed sludge was added.

a total of three times and allowed to drain. The wet, pretreated biomass was then transferred to the digester.

Because switchgrass performed well using similar pretreatment in our prior research, it was not included in these batch studies.

Batch studies

Short-term batch studies were used to screen pretreatment conditions. One-liter flasks were filled with 500 mL seed sludge plus 10 g (dry weight) of pretreated biomass. The bench-scale digester setup is pictured in Figure 2. Water baths were controlled at $39^{\circ}\text{C} \pm 1^{\circ}\text{C}$ ($105^{\circ}\text{F} \pm 2^{\circ}\text{F}$) and digesters were mixed intermittently (10 s each day by manually swirling the reactors). Biogas generated from each digester was collected in a graduated cylinder and displaced the water inside the cylinder to a water reservoir. The reservoir was placed lower than the cylinder to maintain a siphon and produce slight negative pressure in the airspace of the cylinder and digester. Biogas production volume was measured daily by the change in cylinder water level (after placing the reservoir and cylinder at the same level to eliminate pressure differences). To analyze for methane concentration in the biogas, the water reservoir was raised above the cylinder to force the biogas out through a take-off valve connected to the methane analyzer.

Cumulative methane production was then measured over the next 9 days without additional feeding. All conditions were run in triplicate. Each time a new pretreatment condition was evaluated; a new set of control digesters was established to adjust for variations in seed culture activity.

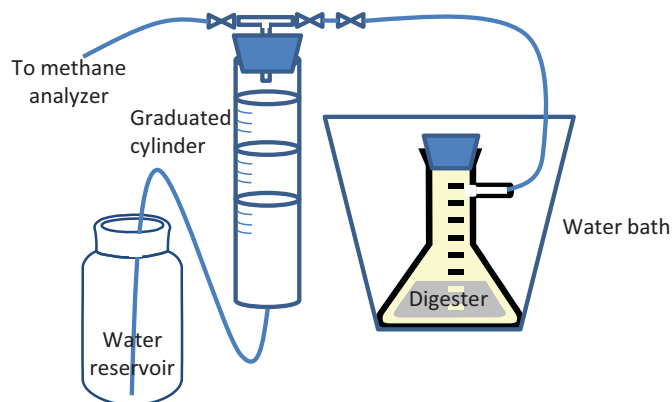


Fig. 2. Bench-top digester and gas collection/measurement setup.

Continuous-feed studies

The most promising pretreatment conditions from the batch studies were evaluated using longer-term (approximately 8 weeks) continuously fed digesters in order to assess potential long-term operational concerns such as solids accumulation, poor mixing, or floating layers. It also allowed verification of digestion under steady-state conditions. One-liter digesters were operated to simulate a typical US dairy-based plug-flow digester.

Switchgrass was included in the continuous-feed studies using pretreatments found to be most effective in our previous study of municipal wastewater treatment AD: fine-grinding, and coarse-grinding followed with 6.0% NaOH-heated pretreatment. Similar to our previous study, biomass loading rates were 4% weight/volume of feed sludge (i.e., raw manure) resulting in a feed total solid (TS) of 12.48% solids and organic loading rate of $3.27 \text{ kg VS m}^{-3} \text{ d}^{-1}$ (31% of VS was from switchgrass).

The digester and gas collection setup was essentially the same as in batch studies (see Fig. 2) but digesters were fed every-other day, following withdrawal of a 30-mL sludge sample. The mixing was intermittent (10 s each day by manually swirling the reactors). To withdraw a sludge sample, the digester was disconnected from the graduated cylinder and connected to a nitrogen cylinder through the side-arm tubing (see Fig. 3). Nitrogen gas was then introduced and the flask stopper was replaced with a stopper/tube assembly. Nitrogen pressure created inside the digester pushed the sludge through the sampling tube into a collection flask. Sludge was stirred during withdrawal process to assure representativeness of the sample. The stopper/tube assembly was then removed and 30 mL of feed sludge/biomass mix was poured into the flask while the nitrogen purge continued. Feed sludge was raw manure. Controls received feed sludge without biomass. All conditions were run in triplicate. Digester performance was monitored as methane production, total solids, volatile solids, and pH. pH was self-sustained, no adjustment was needed. Pretreatment effect was evaluated using data collected only after digesters reached equilibrium (750 h).

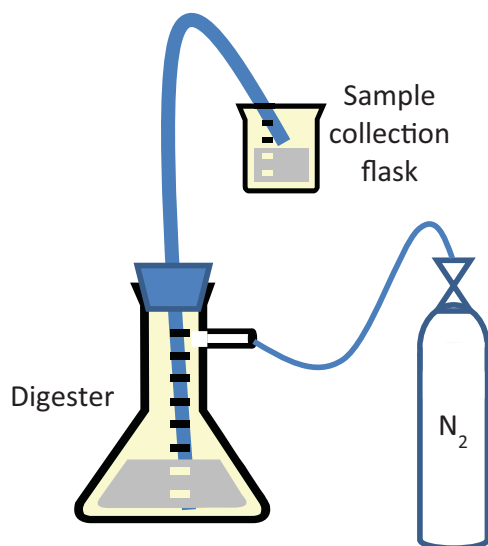


Fig. 3. Anaerobic sludge sampling setup.

Analysis and calculations

Methane analysis

Methane analysis was conducted using Hegmyer Model MX6 -0000N201 iBrid™ methane analyzer from Industrial Scientific Corporation (Oakdale, PA, USA) with a detection limit of 1% and a sensitivity of $\pm 1\%$. Calibration of the instrument was conducted at the beginning and end of each lab session to adjust for any instrument drift during use.

Sludge analysis

For the continuous-feed study, pH of the sludge was measured using a Fisher Scientific pH meter (Vernon Hills, IL, USA), and TS and VS were analyzed according to Standard Method 2540-B and Standard Method 2540-E, respectively.

Digestibility in the batch studies

To assess the digestibility of biomass in the batch studies, cumulative methane production under each condition was compared to the control (i.e., no biomass added) during the same digestion period. The increase is the methane production contributed by biomass. To evaluate if the methane increase is statistically significant, *t*-tests were conducted and *P*-values are reported. In addition, energy conversion efficiency (%) is also calculated (details of calculation can be found in the last paragraph of this section).

Digestibility in continuous-feed studies

To access the digestibility effect in the continuous-feed studies, specific methane yield, energy conversion

efficiency, and other operational parameters are reported for each condition. The calculation of specific methane yield and energy conversion efficiency is described below.

Specific methane yield

The methane generated under each condition was expressed as $\text{m}^3 \text{kg}^{-1} \text{BM}_{\text{added}}$ (BM stands for biomass) and $\text{m}^3 \text{kg}^{-1} \text{VS}_{\text{added}}$. The latter was calculated based on the VS of biomass added to the digester. These values were estimated by fitting a regression line to the data (daily BM or VS added, and daily methane production) for each condition and the controls. The slope of the regression line represents the specific methane yield.

The statistical significance of each slope term was evaluated using a modified ANOVA process to account for the lack of independence of observations on a given digester and that each digester was used in only one condition. Sums of squares were calculated for within-digester, between-digester (but within-treatment group), and between-treatment groups. An *F*-statistic was calculated for the slope terms from mean squares as $\text{MS}_{\text{between-treatment}}/\text{MS}_{\text{within-digester}}$ and degrees of freedom of 1, $n-r$, where n is the number of observations and r is the number of digesters. All statistical analyses were performed with PASW Statistics 18.

Gross heat of combustion

The gross heat of combustion of biomass used in this study was determined using an oxygen bomb calorimeter (IKA Model C2000) at initial oxygen pressure of 3.0 MPa (30 atm) and final temperature of 20–30°C. Benzoic acid was used for calibration. Gelatin caps were used to hold biomass in place for ignition.

Energy conversion efficiency (%)

Energy conversion efficiency is calculated based on gross heat of combustion of added biomass and heat content of methane generated from biomass in the digesters. Energy conversion efficiency was therefore expressed as percentage of biomass energy content converted into methane.

Results and discussion

Batch studies

Cumulative methane production under the various pretreatment conditions for maple leaves, oak leaves, and corn stover, as compared to controls (no biomass added), is summarized in Tables 4, 5, and 6, respectively.

Results for maple and oak leaves (Tables 4 and 5) suggest that these leaves are not easily digested even with the

Table 4. Specific methane yield and energy conversion for maple leaves in batch digestion.

<i>Pretreatment conditions</i>	<i>Specific methane yield $\pm$ Standard deviation ($L\ kg^{-1}$ of BM_{added})</i>	<i>Energy conversion efficiency^a</i>	<i>P-value</i>
0% NaOH, 25°C, 18 h, no rinse	97 $\pm$ 32	20.7%	0.108
3% NaOH, heat for 6 h, no rinse	103 $\pm$ 32	22%	0.103
6% NaOH, heat for 24 h, no rinse	0	NA	NA
15% NaOH, heat for 24 h, no rinse	0	NA	NA
15% NaOH, heat for 24 h, rinse	42 $\pm$ 21	9.1%	0.371
6% NaOH, autoclave, no rinse	23 $\pm$ 16	4.9%	0.634
15% NaOH, autoclave, no rinse	0	NA	NA
15% NaOH, autoclave, rinse	46 $\pm$ 14	9.8%	0.137

^aCalculation can be found in methodology, gross heat content of maple leaf is 7,236 btu lb⁻¹.

pretreatments evaluated in this study. The best digestion was observed for maple leaves pretreated with RO water at room temperature (condition A, Table 4) and with 3% NaOH at 100°C (condition B, Table 4), achieving about 20% energy conversion efficiency, but results were not statistically significant. Without rinse, pretreatment using higher concentrations of NaOH (such as conditions C, D, and G in Table 4; and condition C in Table 5) resulted in no methane yields. This suggests NaOH at higher doses inhibits anaerobic digestion, which could be due to higher pH (not measured) or production of inhibitory compound during the pretreatment. All the other pretreatment conditions for maple leaves resulted in less than 10% energy conversion.

Results in Table 5 suggest that oak is even more difficult to digest than maple. The most effective pretreatment conditions for maple (RO water or 3% NaOH) had essentially no effect on oak. Even fine grinding – a pretreatment unlikely to be economically feasible – resulted in only a small improvement in methane production and energy conversion efficiency (condition D). Adding NaOH to fine grinding produced no improvement and may even have inhibited digestion (conditions E and F).

Overall, these results suggest that the pretreatments evaluated in this study – NaOH, mild heat, autoclaving, and fine grinding – are unlikely to improve digestion of maple and oak leaves to the extent that they could be successfully codigested with manure in liquid-state anaerobic digesters.

In contrast, corn leaves demonstrated relatively good digestion whether pretreated with NaOH under conditions of mild heat or autoclaving (conditions A and B, Table 6) with energy conversion efficiencies in excess of 38% and *P*-values under 0.01. Corn stalk appears somewhat more resistant than corn leaf but still demonstrated moderately good digestion (over 27% energy conversion efficiency and *P*-values under 0.01).

Continuous-feed studies

In order to assess potential long-term operational concerns typically associated with digestion of lignocellulosic feedstocks such as solids accumulation, poor mixing, or floating layers, several conditions were evaluated using 8-week continuously feed bench-top digesters. From the batch study, only corn leaves and corn stalks were able to digest well. Therefore, corn stover (a mix of corn leaf and corn stalk at 1-to-1 ratio) with the most promising pretreatment condition identified from the batch study (i.e., 3% NaOH, heated, no rinse) was tested using continuous-feed digesters. Switchgrass pretreatments reflected the pretreatments found to be most effective in our previous study of municipal wastewater treatment AD: fine-grinding, and coarse-grind treated with 6% NaOH.^[23]

The trend over time for total solids in the digesters demonstrated a slow increase from 5.0%–5.5% up to 5.0%–6.0% within about one retention period (30 days) followed

Table 5. Specific methane yield and energy conversion for oak leaves in batch digestion.

<i>Pretreatment conditions</i>	<i>Specific methane yield $\pm$ Standard deviation ($L\ kg^{-1}$ of BM_{added})</i>	<i>Energy conversion efficiency</i>	<i>P-value</i>
0% NaOH, 25°C, 18 h, no rinse	9 $\pm$ 6	1.9%	0.543
3% NaOH, heat for 24 h, no rinse	11 $\pm$ 7	2.3%	0.562
3% NaOH, autoclave, no rinse	0	NA	NA
D) fine grind	68 $\pm$ 12	14.6%	0.033
Fine grind, 6% NaOH, 25°C for 18 h, no rinse	43 $\pm$ 18	9.2%	0.132
Fine grind, 6% NaOH, heat for 24 h, no rinse	51 $\pm$ 10	10.9%	0.075

^aCalculation can be found in methodology, gross heat content of oak leaf is 7,304 btu lb⁻¹.

Table 6. Specific methane yield and energy conversion for corn leaves and stalks in batch digestion.

<i>Pretreatment conditions</i>	<i>Specific methane yield $\pm$ Standard deviation ($L\ kg^{-1}$ of BM_{added})</i>	<i>Energy conversion efficiency</i>	<i>P-value</i>
Corn leaf, 3% NaOH, 6 h 100°C, no rinse	180 $\pm$ 19	38.6%	0.002
Corn leaf, 3% NaOH, autoclave 1 h, no rinse	183 $\pm$ 19	39.3%	0.001
Corn stalk, 4% NaOH, 6 h 100°C, no rinse	134 $\pm$ 12	28.8%	0.0001
Corn stalk, 4% NaOH, autoclave 1 h, no rinse NaOH, overnight	126 $\pm$ 13	27.1%	0.003

^aCalculation can be found in methodology, gross heat content of corn leaf and corn stalk are 7,214 and 7,162 btu lb⁻¹, respectively.

by a leveling off of the solids concentration (data not shown). This suggests that the digesters had reached equilibrium by about 30 days, and that solid levels are not likely to be high enough to cause operational problems for typical US farm digesters (operating at 5.0%–6.0% solids). In addition, problems with mixing and/or floating material were not observed in any of the tested conditions.

Figures 4a–f present cumulative methane production by pretreatment condition for each of the triplicate test digesters and triplicate control digesters used for comparison (made from the same seed culture as the test digesters). Data indicate relatively good agreement among triplicates and a clear distinction between test and control digesters in each condition. Methane production was relatively stable, particularly after 750 h, used as the starting point of steady-state conditions.

Specific methane yield, energy conversion efficiency, and other operational and performance parameters for all conditions are presented in Table 7. Note that VS_{added} includes only volatile solids from added biomass, not from raw manure or seed culture. In contrast, VS removal is based on total volatile solids reduction from the combination of seed culture, raw manure, and added biomass. Energy conversion efficiency refers only to additional methane from added biomass in comparison to the energy content of that biomass.

Based on VS removal, all pretreatment conditions demonstrated good digestion, ranging from 56% to 67% removal. However, methane production and energy conversion was much higher with switchgrass than corn stover. Codigestion of corn stover demonstrated a high level of VS removal but lower concentration of methane in the biogas, as compared to coarse-grind switchgrass. Fine-grind switchgrass also demonstrated a relatively low concentration of methane in biogas while maintaining an overall high level of VS removal. Specific methane yields ranged from 206 to 348 $L\ kg^{-1}\ VS_{added}$.

The current findings for switchgrass contrast dramatically with our previous work using seed sludge from a municipal wastewater biosolids digester.^[23] In batch studies using municipal seed sludge, pretreatment of coarse-grind switchgrass with 6% NaOH increased specific methane yield more than 20-fold in comparison to untreated switchgrass. In contrast, the current study using dairy digester

seed sludge found that untreated and 6% NaOH pretreated switchgrass had very similar specific methane yields, comparable to 6% NaOH pretreated switchgrass using municipal seed sludge. The most likely explanation for this finding is the acclimation of dairy digester micro-organisms to undigested lignocellulosic material in dairy cattle manure, allowing untreated switchgrass to be readily digested.

Comparison to other research

Our results for oak and maple leaves are somewhat lower than prior research on apple and peach leaves. Hassan Dar and Tandon^[24] pretreated apple and peach leaf litter with a 1% NaOH solution for 7 days at room temperature (equivalent to 10% NaOH wt/wt) and codigested with cattle manure. Pretreated leaves produced about a 35% improvement in methane production compared to cattle manure alone (the digestibility of leaves without pretreatment was not tested). This is generally consistent with the 27%–29% improvement observed in our study when comparing maple leaves pretreated with 0% or 3% (wt/wt) NaOH to digestion of manure alone. However, the 10% NaOH used by Hassan Dar and Tandon is in the concentration range that seemed to have an inhibitory effect in our study. It appears that Hassan Dar and Tandon did not rinse the biomass after pretreatment. It is unclear whether treatment at room temperature may have reduced the production of inhibitory compounds in the Hassan Dar and Tandon study, or oak and maple leaves may have a greater tendency to produce inhibitory compounds when pretreated with NaOH.

Our results are generally consistent with the findings of Brown et al.^[11] Leaf litter (species not identified) and corn stover were digested without codigestion using effluent from a municipal wastewater treatment digester. Feedstocks were not pretreated other than grinding to a size that would pass a 5-mm screen. While direct comparison of methane yield between our results and theirs is not possible due to differences in the period of gas collection, they found the methane yield from leaves to be about 60% of that from corn stover, similar to our results when comparing maple leaves pretreated with 0% or 3% (wt/wt) NaOH to pretreated corn leaves and stalks.

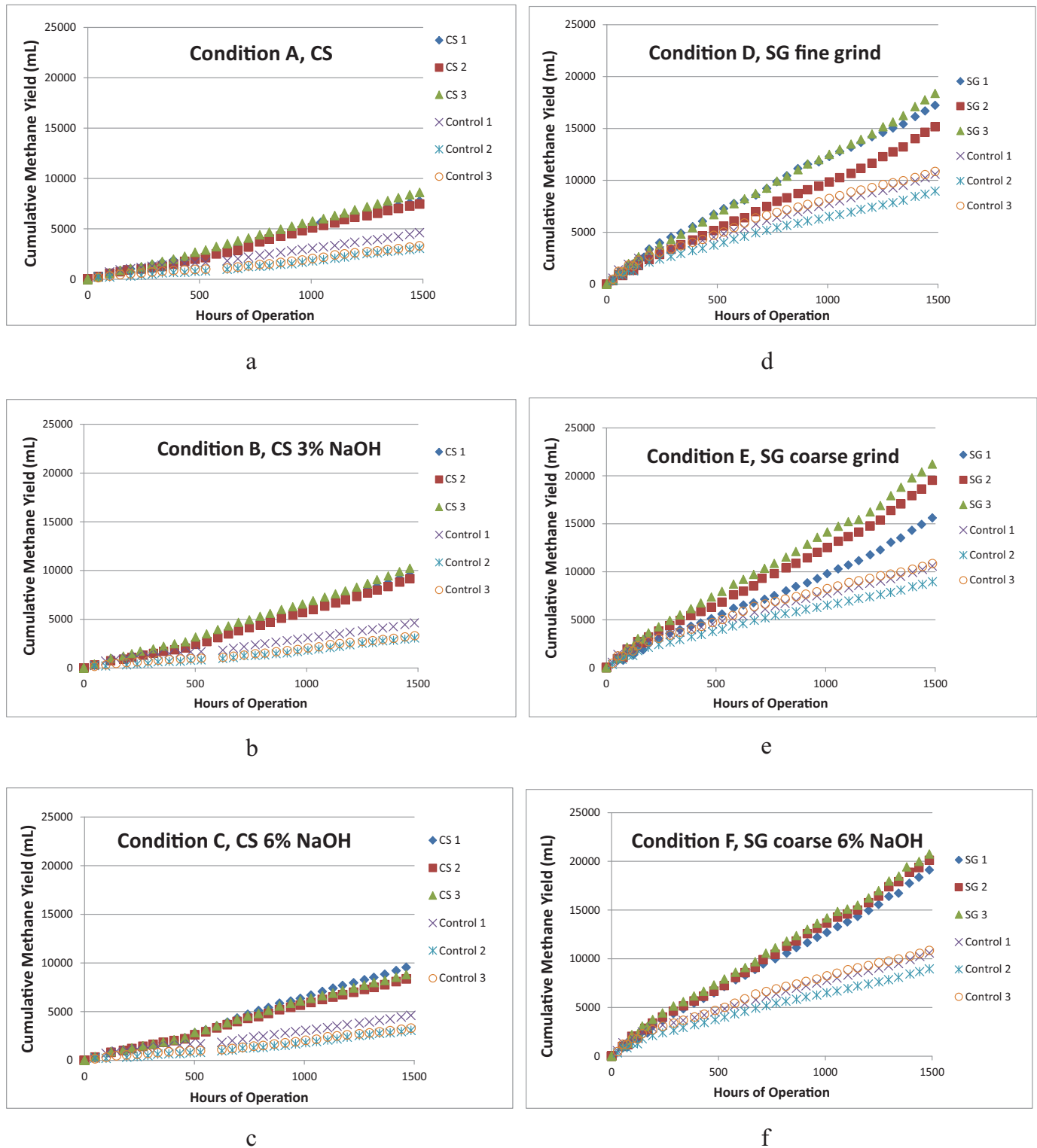


Fig. 4. (a–f) Cumulative methane production in 8-week continuously feed digestion of corn stover (CS) and switchgrass (SG) by pre-treatment condition (triplicate digesters indicated by number). (a) CS only; (b) CS with 3% NaOH; (c) CS with 6% NaOH; (d) SG fine-grind; (e) SG coarse grind; (f) SG coarse grind with 6% NaOH.

Table 7. Results for 8-week continuously feed digestion of corn stover (CS) and switchgrass (SG) by pretreatment condition (data from over 750 h of digestion only).

Pretreatment conditions ^a	Total VS removal ^b	% of Biogas that is methane	Specific methane yield per unit biomass ($L\ kg^{-1}\ BM_{added}$)	Specific methane yield per unit VS_{added} ($L\ kg^{-1}\ VS_{added}$) ^c	Energy conversion efficiency ^d
CS	61.5%	42%	182*	206*	39%
CS, 3% NaOH	67.3%	43%	201*	227*	43%
CS 6% NaOH	64.6%	45%	191*	216*	41%
SG fine grind	56.2%	48%	222*	252*	48%
SG coarse grind	55.7%	58%	289*	328*	62%
SG coarse grind, 6% NaOH	58.1%	63%	307*	348*	66%

^aFeed TS are 12.30% for corn stover and 12.48% for switchgrass, respectively; organic loading rates ($kg\ VS\ m^{-3}\ d^{-1}$) are 3.24 for corn stover and 3.27 for switchgrass, respectively; % feed VS from biomass (wt/wt) are 31.73% from corn stover and 32.41% for switchgrass, respectively.

^bTotal VS removal is calculated based on total volatile solids reduction from farm digester feed sludge and switchgrass/corn stover.

^cVS here is only volatile solids from corn stover or switchgrass, not including those from farm digester feed sludge.

^dCalculations can be found in methodology. Gross heat contents of switchgrass and corn stover are 7,459 and 7,186 $btu\ lb^{-1}$, respectively.

*Statistically significant at $P < 0.01$.

Liew et al.^[25] examined the effect of simultaneous dry digestion and treatment with NaOH on leaf litter (species not identified) without codigestion. Various NaOH loading rates and substrate/inoculum (S/I) ratios were examined in batch studies. Overall, methane yield improvements were greatest at about 3.5% NaOH, with inhibition occurring as NaOH increased above that concentration.

Results for corn stover are roughly comparable to those of Xiao et al.^[26] Corn stover was pretreated with either NaOH or a combination of NaOH/Ca(OH)₂ for 3 days at 20°C, and was not rinsed prior to use. In batch studies without codigestion, VS removal ranged from 54% to 74%. The specific methane yields of 269–287 $L\ kg^{-1}\ VS_{added}$ (20-day collection period) and are higher than the 126–183 $L\ kg^{-1}\ VS_{added}$ found in our batch studies but are comparable to the 206–348 $L\ kg^{-1}\ VS_{added}$ found in our continuous-feed studies. Xiao et al.^[26] found that at smaller particle size (0.25–1.0 mm), alkaline pretreatment only improved methane yield about 15% compared to untreated stover of the same particle size. At a particle size of 5.0–20.0 mm, pretreatment produced an improvement of 46%–48%.

Pang et al.^[27] pretreated corn stover of 5–10 mm particle size with NaOH concentrations ranging from 4% to 10% (wt/wt). Biomass was pretreated for 3 weeks at 20°C. Treated stover was apparently unrinsed prior to digestion in batch digesters without codigestion, but ammonium chloride was added to achieve a more optimal C:N ratio. At the lowest biomass loading rate (35 $g\ L^{-1}$), pretreatment had relatively little effect, improving methane production only 14%–24% over untreated stover. As loading rates increased to a maximum of 80 $g\ L^{-1}$, however, pretreatment effect increased steadily to a maximum of 80%–91% more methane than untreated stover. All NaOH concentrations produced relatively similar improvements, but the optimal concentration seemed to be about 6%–8% with some inhibition occurring at 10%.

Zhu et al.^[28] pretreated corn stover with NaOH at concentrations from 1% to 7.5% (wt/wt) for 1 day at ambient temperature prior to solid-state anaerobic digestion. They found that optimal NaOH concentration was 5%, producing an increase of 37% in methane yield (372 $L\ kg^{-1}\ VS_{added}$) compared to untreated stover. At 7.5% NaOH, methane production was inhibited. These results are comparable to our batch study results indicating inhibition at concentrations higher than 5% NaOH.

Zheng et al.^[29] pretreated corn stover with 2% NaOH (wt/wt), temperatures (10–50°C), and times (3–21 days) prior to batch digestion without codigestion. Stover had been ground to 5–10 mm and treated stover was apparently unrinsed prior to digestion. Results demonstrated little improvement in methane production resulting from increasing temperature or treatment period. In fact, the highest temperatures and longest treatment time resulted in decreasing methane production. Optimal conditions, among those tested, were 3-day treatment time at 20°C, producing a methane yield of 233 $L\ kg^{-1}\ VS_{added}$.

Labatut and Scott^[30] found that non-pretreated switchgrass actually reduced methane production when mixed in a 1:3 ratio with manure, as compared to manure alone. They noted that the switchgrass was “mixed and blended to reduce particle size” but did not note the final particle size. Ahn et al.^[31] pretreated switchgrass by fine-grinding to approximately 2 mm and codigesting with swine manure, dairy manure, or poultry manure at 15% solids in a batch-digestion process. Both dairy and poultry manure failed to sustain anaerobic digestion due to excess acid production. However, swine manure had sufficient buffering capacity to sustain digestion, resulting in a methane yield of 337 $L\ kg^{-1}\ VS_{added}$, roughly comparable to our findings using 6% NaOH.

Himmelsbach et al.^[32] pretreated switchgrass in 29.5% aqueous ammonium (5 $mL\ g^{-1}$ of switchgrass, equivalent to 148% wt/wt alkaline/biomass) for 5 days at room temperature. In 21-day batch digestion studies, ammonia

pretreatment resulted in a methane yield of about 160 L kg⁻¹ VS, an increase of about 65% compared to untreated switchgrass. However, this methane production represents only about 30% of theoretical maximum, in contrast to the 66% energy yield observed in the current study using 6% NaOH.

These results support the proposition that alkali concentration, treatment time, and treatment temperature act together in producing the overall pretreatment effect. Reductions in one or two of these factors can be offset, at least in part, by increases in the other factors. But increases in these factors ultimately lead to decreases in methane production, either through destruction of biodegradable compounds, production of inhibitory chemicals, or some other mechanism.

Conclusion

This research, in combination with prior studies of leaves, corn stover, and switchgrass digestion, suggests that mild heating and relatively low concentrations of NaOH can be effective in improving methane yield from these feedstocks. However, leaves from oak and maple, common urban tree species, are unlikely to digest well enough in typical farm-based digesters to support their use as a feedstock. Alternative pretreatments will be required before yard waste could become an important feedstock in plug-flow or other “liquid state” digesters.

Results also suggest that the combination of NaOH concentration, treatment time, and treatment temperature probably all play a role in determining the effectiveness of alkaline pretreatment. Selecting the optimal combination is important, since increasing one or more of these factors can ultimately lead to digestion inhibition. However, the impact and optimal combination of these factors is probably dependent not only on the type of feedstock being treated but also loading rate to the digester.

Feedstock digestion, as measured by VS removal, does not always correlate with high specific methane yield. Corn stover and fine-grind switchgrass demonstrated high VS removal but comparatively low methane yield. These results suggest that hydrolysis and fermentation of these feedstocks may have proceeded more quickly than methanogenesis, though not to the point of suppressing pH. Reducing the loading of these readily digested feedstocks and blending with somewhat more recalcitrant feedstocks may avoid the problem of excess CO₂ production.

One factor that appeared to play an important role in our results, but has received comparatively little attention in the literature, is the role of microbial acclimation. In our research, the use of dairy-based seed sludge, which had likely acclimated to the high lignocellulosic content of manure, resulted in methane yield from untreated switchgrass that was over 20-fold higher than when municipal seed sludge was used. These results suggest that

acclimation could be an important tool in successful digestion of urban yard waste and other recalcitrant feedstocks in farm-based digesters.

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References

- [1] Moser, M.A. Anaerobic Digesters Control Odors, Reduce Pathogens, Improve Nutrient Manageability, Can be Cost Competitive with Lagoons, and Provide Energy Too. In *Proceedings from Manure Management in Harmony with the Environment and Society Conference. Soil and Water Conservation Society*, Feb 10–12, 1998; Iowa State University: Ames, IA, 1998. Available at http://www.epa.gov/agstar/documents/lib-man_man.pdf (accessed Apr 2014).
- [2] Ward, A.J.; Hobbs, P.J.; Holliman, P.J.; Jones, D.L. Optimisation of the anaerobic digestion of agricultural resources. *Biores. Technol.* **2008**, *99*(17), 7928–7940.
- [3] US Environmental Protection Agency (US EPA). *Operating Anaerobic Digester Projects*. Cincinnati, OH, Author, 2014. Available at <http://epa.gov/agstar/projects/index.html> (accessed Apr 2014).
- [4] Gloy, B.A.; Dressler, J.B. Financial barriers to the adoption of anaerobic digestion on US livestock operations. *Agric. Financ. Rev.* **2010**, *70*(2), 157–168.
- [5] Center for Climate and Energy Solutions (C2ES). *Anaerobic Digesters Quick Facts*; Center for Climate and Energy Solutions, Arlington, VA, 2011. Available at <http://www.c2es.org/docUploads/AnaerobicDigesters.pdf> (accessed Apr 2014).
- [6] van Forest, F. *Perspectives for Biogas in Europe*; Oxford Institute for Energy Studies, Oxford, United Kingdom, 2012. Available at <http://www.oxfordenergy.org/wpcms/wp-content/uploads/2012/12/NG-70.pdf> (accessed Apr 2014).
- [7] Wang, X.; Yang, G.; Feng, Y.; Ren, G.; Han, X. Optimizing feeding composition and carbon–nitrogen ratios for improved methane yield during anaerobic co-digestion of dairy, chicken manure and wheat straw. *Biores. Technol.* **2012**, *120*, 78–83.
- [8] US EPA. *Increasing Anaerobic Digester Performance with Codigestion*; Cincinnati, OH, Author, 2012. Available at <http://www.epa.gov/agstar/documents/codigestion.pdf> (accessed Apr 2014).
- [9] Weiland, P. Biogas production: current state and perspectives. *Appl. Microbiol. Biot.* **2010**, *85*(4), 849–860.
- [10] US EPA. *U.S. Anaerobic Digester Status Report*; Cincinnati, OH, Author, 2010. Available at http://www.epa.gov/agstar/documents/digester_status_report2010.pdf (accessed Apr 2014).
- [11] Brown, D.; Shi, J.; Li, Y. Comparison of solid-state to liquid anaerobic digestion of lignocellulosic feedstocks for biogas production. *Bioresour. Technol.* **2012**, *124*, 379–386.
- [12] US EPA. *Municipal Solid Waste Generation, Recycling, and Disposal in the United States, Facts and Figures for 2012*; Cincinnati, OH, Author, 2012. Available at http://www.epa.gov/waste/nonhaz/municipal/pubs/2012_msw_fs.pdf (accessed Apr 2014).
- [13] Gallagher, P.; Baumes, H. *Biomass Supply from Corn Residues: Estimates and Critical Reviews of Procedures*. Agricultural Economic Report Number 847. U.S. Department of Agriculture,

- Washington, DC, 2012. Available at <http://www.usda.gov/oce/reports/energy/Biomass%20Supply%20From%20Corn%20Residue-Nov-2012.pdf> (accessed Apr 2014).
- [14] Thompson, J.; Tyner, W. *Corn Stover for Bioenergy Production: Cost Estimates and Farmer Supply Response*; Purdue University Extension, West Lafayette, IN, 2011. Available at www.extension.purdue.edu/extmedia/EC/RE-3-W.pdf (accessed Apr 2014).
- [15] Downing, M.; Eaton, L.M.; Graham, R.L.; Langholtz, M.H.; Perlack, R.D.; Turhollow Jr, A.F.; Brandt, C.C. *US Billion-Ton Update: Biomass Supply for a Bioenergy and Bioproducts Industry*; Oak Ridge National Laboratory (ORNL), U.S. Department of Energy: Washington, DC, 2011; No. ORNL/TM-2011/224, 87–144. Available at <http://info.ornl.gov/sites/publications/files/Pub31057.pdf> (accessed Apr 2014).
- [16] Perrin, R.; Vogel, K.; Schmer, M.; Mitchell, R. Farm-scale production cost of switchgrass for biomass. *Bio Ener. Res.* **2008**, *1*(1), 91–97.
- [17] Zhang, P.F.; Zhang, Q.; Pei, Z.J.; Wang, D.H. Cost estimates of cellulosic ethanol production: a review. *J. Manuf. Sci. Eng.* **2013**, *135*(2), 021005.
- [18] Kumar, P.; Barrett, D.M.; Delwiche, M.J.; Stroeve, P. Methods for pretreatment of lignocellulosic biomass for efficient hydrolysis and biofuel production. *Ind. Eng. Chem. Res.* **2009**, *48*, 3713–3729.
- [19] Frigon, J.C.; Guiot, S.R. Biomethane production from starch and lignocellulosic crops: a comparative review. *Biofuels, Bioprod. Biorefin.* **2010**, *4*(4), 447–458.
- [20] Taherzadeh, M.J.; Jeihanipour, A. Recalcitrance of lignocellulosic biomass to anaerobic digestion. In *Pretreatments for Enhanced Anaerobic Technology*; Mudhoo, A., Ed.; Co-published by John Wiley & Sons, Inc. Hoboken, New Jersey, USA, and Scrivener Publishing LLC, Salem, MA, 2012; 27–54.
- [21] Brodeur, G.; Yau, E.; Badal, K.; Colier, J.; Ramachandran, K.B.; Ramakrishnan, S. Chemical and physicochemical pretreatment of lignocellulosic biomass: a review. *Enzyme Res.* **2011**. doi:10.4061/2011/787532 <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3112529/> (accessed July 2014)
- [22] Hendriks, A.T.; Zeeman, G. Pretreatments to enhance digestibility of lignocellulosic biomass. *Biores. Technol.* **2009**, *100*(1), 10–18.
- [23] Jin, G.; Bierma, T.; Walker, P.M. Low-heat, mild alkaline pretreatment of switchgrass for anaerobic digestion. *J. Environ. Sci. Health, Part A* **2014**, *49*(5), 565–574.
- [24] Hassan Dar, G.; Tandon, S.M. Biogas production from pretreated wheat straw, lantana residue, apple and peach leaf litter with cattle dung. *Biolog. Wastes* **1987**, *21*(2), 75–83.
- [25] Liew, L.N.; Shi, J.; Li, Y. Enhancing the solid-state anaerobic digestion of fallen leaves through simultaneous alkaline treatment. *Biores. Technol.* **2011**, *102*(19), 8828–8834.
- [26] Xiao, X.; Zhang, R.; He, Y.; Li, Y.; Feng, L.; Chen, C.; Liu, G. Influence of particle size and alkaline pretreatment on the anaerobic digestion of corn stover. *BioResources* **2013**, *8*(4), 5850–5860.
- [27] Pang, Y.Z.; Liu, Y.P.; Li, X.J.; Wang, K.S.; Yuan, H.R. Improving biodegradability and biogas production of corn stover through sodium hydroxide solid state pretreatment. *Energy Fuels* **2008**, *22*(4), 2761–2766.
- [28] Zhu, J.; Wan, C.; Li, Y. Enhanced solid-state anaerobic digestion of corn stover by alkaline pretreatment. *Biores. Technol.* **2010**, *101*(19), 7523–7528.
- [29] Zheng, M.; Li, L.; Li, X.; Xiong, J.; Mei, T.; Chen, G. The effects of alkaline pretreatment parameters on anaerobic biogasification of corn stover. *Energy Source, Part A: Recovery, Utilization, and Environmental Effects* **2010**, *32*, 1918–1925.
- [30] Labatut, R.A.; Scott, N.R. Experimental and predicted methane yields from the anaerobic co-digestion of animal manure with complex organic substrates. In *Proceedings of the American Society of Agricultural and Biological Engineers (ASABE) Annual International Meeting*, Providence, RI, June 29–July 2, 2008.. ASABE: St. Joseph, MI, 2008; Paper No. 085087, 6940–6957.
- [31] Ahn, H.K.; Smith, M.C.; Kondrad, S.L.; White, J.W. Evaluation of biogas production potential by dry anaerobic digestion of switchgrass–animal manure mixtures. *Appl. Biochem. Biotechnol.* **2010**, *160*(4), 965–975.
- [32] Himmelsbach, J.N.; Raman, D.R.; Anex, R.P.; Burns, R.T.; Faulhaber, C.R. Effect of ammonia soaking pretreatment and enzyme addition on biochemical methane potential of switchgrass. *Trans. ASABE* **2010**, *53*(6), 1921–1928.